Compact 5V USB-powered bridge stereo audio amplifier module based on the TDA2822.

1 Features

- USB powered (5V)
- Bridge-tied load (BTL) stereo amplifier configuration
- Based on the TDA2822 low-voltage audio amplifier IC
- Increased output power using bridge mode operation
- Stereo input via audio jack
- On-board volume control using trim potentiometers
- Power LED indicator for operational status
- Compact PCB design

2 Applications

- USB-powered speaker systems
- Compact desktop audio amplifiers
- Portable audio devices and DIY speakers
- Educational projects and audio electronics demonstrations
- Low-voltage audio amplification experiments
- Prototyping and learning platforms for analog audio circuits
- Embedded systems requiring small audio output stages

3 Description

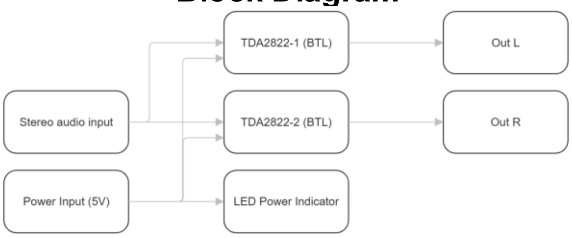
The USB 5V Bridge Stereo Audio Amplifier Module is a compact low-voltage audio amplification solution designed for portable, educational, and embedded applications. The module is based on the TDA2822 audio amplifier IC, a well-known dual low-power amplifier optimized for operation from a single 5V supply.

The amplifier operates in a bridge-tied load (BTL) configuration, in which each audio channel uses two internal amplifier stages driven in opposite phase. This topology effectively doubles the voltage swing applied to the speaker, significantly increasing the available output power compared to single-ended configurations while operating from the same supply voltage. As a result, the module is well suited for driving 4Ω speakers directly without the need for large output coupling capacitors.

Audio input is provided through a stereo jack connector. The incoming left and right audio signals are routed through on-board trim potentiometers, allowing independent volume adjustment before amplification. The conditioned signals are then fed into the TDA2822 inputs, where they are amplified in bridge mode and delivered to the speaker outputs via screw terminal connectors.

The module is powered from a regulated 5V source, which can be supplied either through a USB Type-C connector. The USB interface is used exclusively for power delivery; the data lines are not connected.

Block Diagram



PCB size (without components)
50mm X 50mm

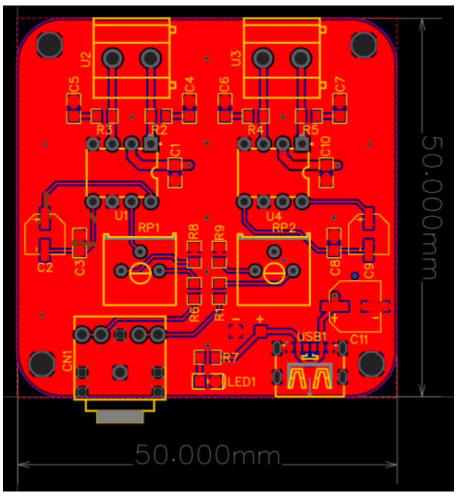


Figure 1. PCB Layout

BOM (Bill of Materials)

Name	Designator	Quantity
0.1u X7R 0805	C1,C4,C5,C6,C7,C10	6
10uF 50V	C2,C9	2
10nF X7R 0805	C3,C8	2
100uF 16V	C11	1
Jack connector	CN1	1
LED-0805 Blue	LED1	1
Resistor 2k2	R1,R6,R8,R9	4
Resistor 4R7	R2,R3,R4,R5	4
Resistor 330R	R7	1
R_3386P_US	RP1,RP2	2
TDA2822	U1,U4	2
KF128-5.0-2P	U2,U3	2
TYPE-C-6P-DIP2X2	USB1	1

Electrical Characteristics

Parameter	Min.	Typ.	Max.	Unit
Supply voltage (VCC)	4.5	5.0	5.5	V
Total quiescent current	-	~25		mA
Output power (per channel, BTL)	-	~0.35		W
Load impedance (recommended)	4	-	8	Ω
Closed-loop voltage gain (TDA2822)	36	39	41	dB
Input resistance (TDA2822)	100	-	-	kΩ
Total harmonic distortion (THD)	-	0.2	-	%
Storage temperature	-40	-	125	°C

* No active short-circuit protection on speaker outputs

Schematic

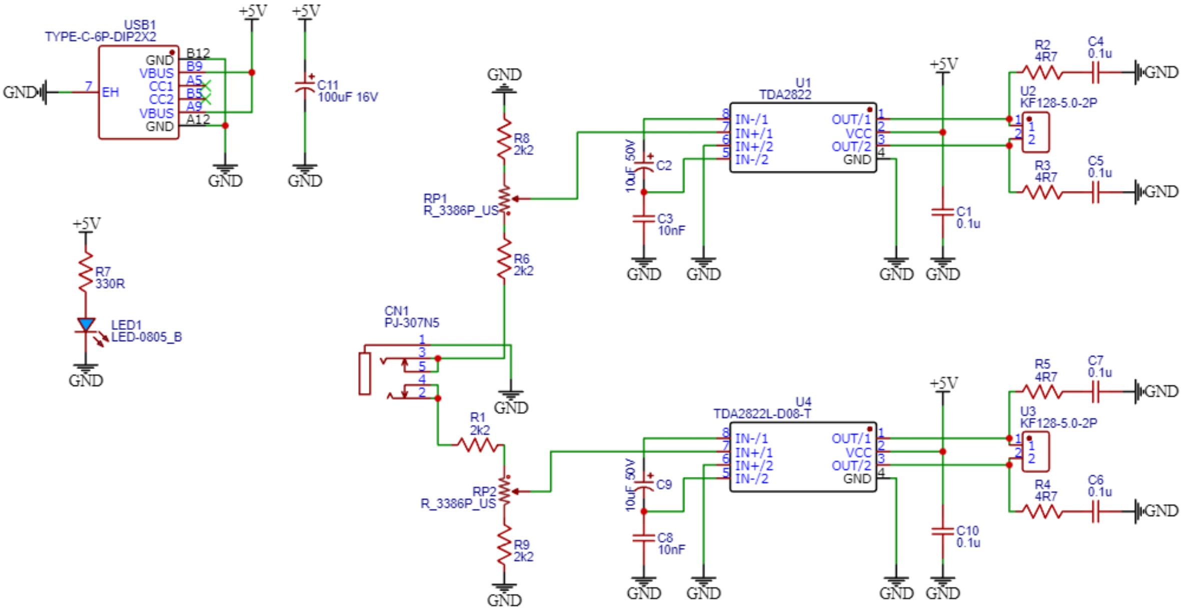


Figure 2. Schematic

3D image of the PCB

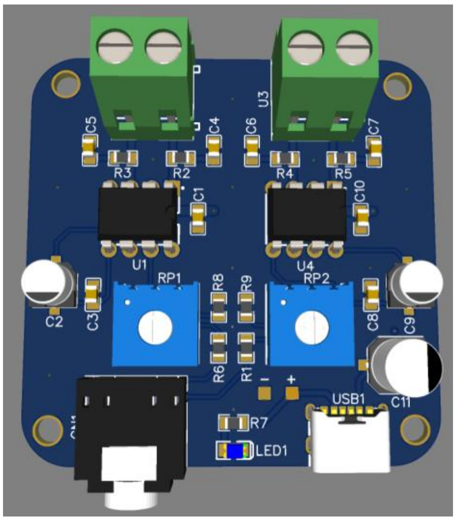


Figure 3. PCB Top Side

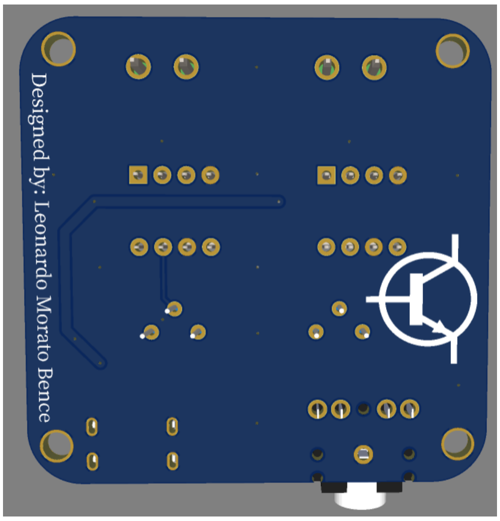


Figure 4. PCB Layout Bottom Side

PCB Assembled

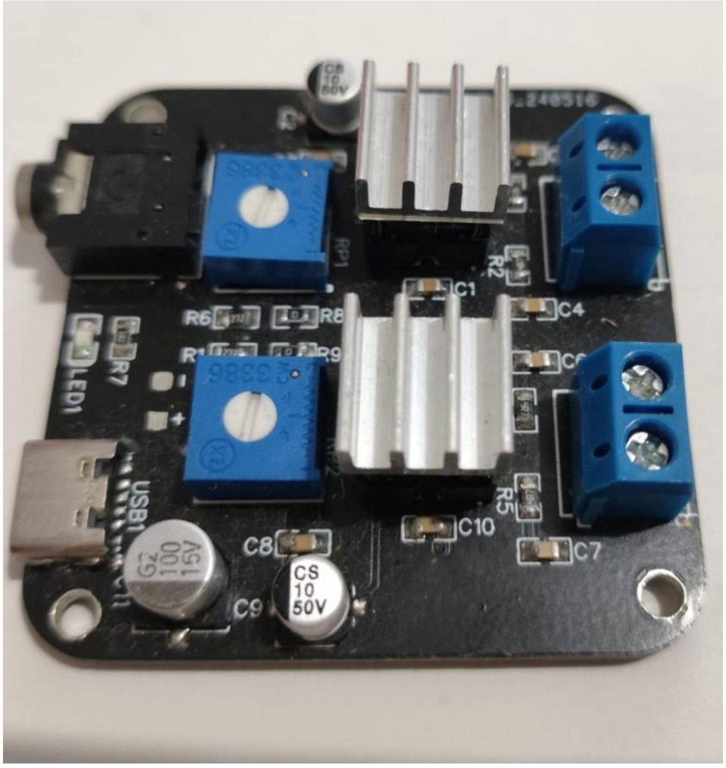


Figure 5. PCB Assembled

Disclaimer

The information contained in this document is provided “as is” and is subject to change without notice. This product is intended solely for educational, prototyping, and demonstration purposes. It is not certified for use in safety-critical, medical, automotive, or industrial applications. The author assumes no responsibility for any damages, direct or indirect, resulting from the improper or incorrect use of this module.

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